

Complex scaling approach to quasinormal modes of Schwarzschild and Reissner–Nordström black holes

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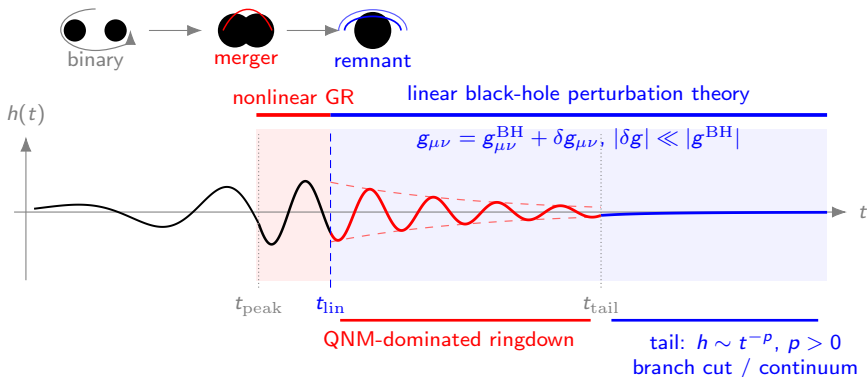
14/9/2026 JPS Meeting @U. of Tokyo

Based on joint work with Shoya Ogawa and Takuya Hirose

- [to appear in PTEP [\[arXiv:2604.20442\]](#)].
- Next talk by Hirose [\[arXiv:2605.03277\]](#) (cf. [\[arXiv:2608.14752\]](#)).

Perturbation theory applies for $t \geq t_{\text{lin}}$

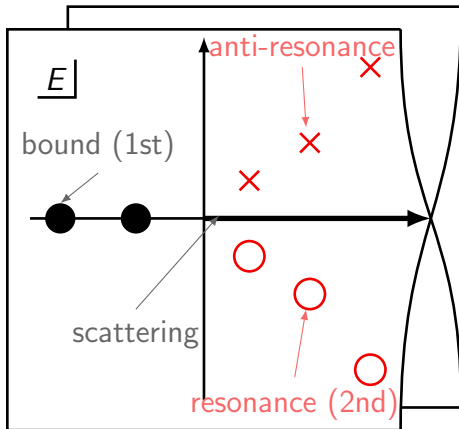
- **Open** black-hole system: ringdown and quasinormal modes (QNMs).



- QNM poles describe the exponentially damped, intermediate-time ringdown. [Price '72; Berti–Cardoso–Starinets '09]

Distribution of bound, resonant, conti states

- Scattering theory for self-adjoint Hamiltonian:
Riemann surface of complex E -plane

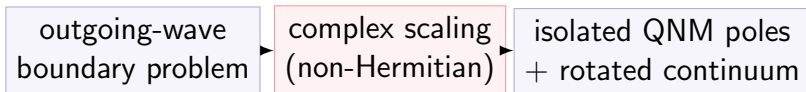


- Complex energy $E = E_r - i\frac{\Gamma}{2}$ in the 4th quadrant
(Resonant energy E_r , decay width Γ)

Why complex scaling for black-hole response?

- QNMs are **resonances**: poles of the analytically continued Green function.
- Continued fractions / exact WKB / time-domain methods can determine poles accurately. [Talks by Miyachi and Shiga]
- But the full response also contains a **non-pole continuum sector**.
- Question: can poles and continuum be represented by **one spectral problem**?

($\rightarrow$ universal class of resolvent singularities in hep / nucl / cond-mat / gr)



- Today: QNMs of Schwarzschild/Reissner–Nordström (OM)
 $\rightarrow$ (dS) Continuum response (Hirose's talk)

Black-hole perturbation as a 1D resonance problem

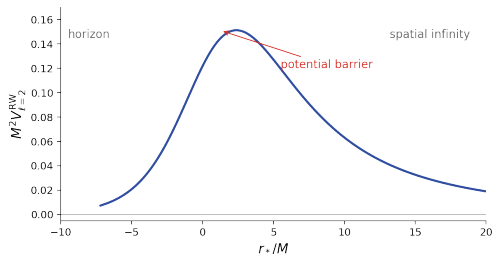
- Schrödinger-type master equation

$$\left[-\frac{d^2}{dr_*^2} + V_\ell(r) \right] \psi_\ell(r_*) = \omega^2 \psi_\ell(r_*).$$

- Radiation conditions

$$\psi_\ell \sim e^{-i\omega r_*} \quad (r_* \rightarrow -\infty), \quad \psi_\ell \sim e^{+i\omega r_*} \quad (r_* \rightarrow +\infty).$$

- The radiation conditions make the spectral problem non-self-adjoint.
- For Schwarzschild odd parity: **Regge–Wheeler barrier**.



Complex scaling regularizes outgoing waves

- Rotate the asymptotic coordinate into the complex plane:

$$r_* \longmapsto r_* e^{i\theta}, \quad 0 < \theta < \frac{\pi}{2}.$$

- For an outgoing wave,

$$\begin{aligned} e^{+i\omega r_* e^{i\theta}} &= e^{-(\operatorname{Re} \omega \sin \theta + \operatorname{Im} \omega \cos \theta) r_*} \\ &\quad \times e^{+i(\operatorname{Re} \omega \cos \theta - \operatorname{Im} \omega \sin \theta) r_*}. \end{aligned}$$

- It decays on the rotated contour when

$$\operatorname{Re} \omega \sin \theta + \operatorname{Im} \omega \cos \theta > 0.$$

- Resonance eigenvalues become discrete and θ -independent; the continuum rotates by 2θ in the $E = \omega^2$ plane.

outgoing boundary condition $\longrightarrow$ non-Hermitian L^2
eigenproblem

[Aguilar–Balslev–Combes theorem]

Finite-basis stability test

- Basis expansion:

$$\psi^\theta(r_*) = \sum_j c_j \phi_j(r_*).$$

- Generalized eigenvalue problem:

$$\sum_j \left(H_{ij}^\theta - \omega^2 N_{ij} \right) c_j = 0.$$

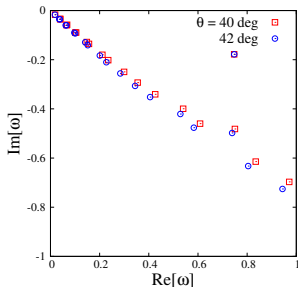
- Vary independently:
 - ▶ the scaling angle θ ;
 - ▶ the number, polynomial order, and geometric range of Gaussians;
 - ▶ the integration range and inverse tortoise map.
- Identify QNMs from isolated stationary branches; use the parameter-variation spread as a numerical stability estimate.
- Broad modes near the rotated continuum require better-balanced bases.

Poles separate from the rotated continuum

- Expand in a polynomial $\times$ real-range Gaussian basis,

$$\phi_{i,n}(r_*) \propto (\sqrt{\alpha_i} r_*)^n e^{-\alpha_i r_*^2},$$

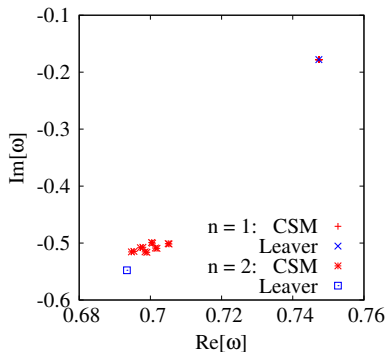
and solve a generalized non-Hermitian eigenvalue problem.



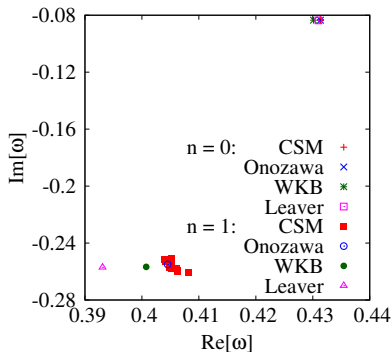
- **QNM candidate**: isolated and stable under changes of θ and basis parameters.
- **Continuum pseudostates**: move with the rotated cut.
- Pseudostates are quadrature data for the continuum, not extra QNMs.

CSM passes Schwarzschild and extremal-RN checks

- The finite-basis CSM reproduces narrow, isolated QNMs without explicit asymptotic Frobenius matching.



Schwarzschild, gravitational, $\ell = 2$



Extremal RN, odd gravitational, $\ell = 2$

- Fundamental Schwarzschild modes agree with Leaver to about 6–7 digits.
- Extremal RN remains accessible when the horizons coalesce and the naive Frobenius structure changes.
- Broad overtones near the rotated continuum are more basis-sensitive.

Summary

- Complex scaling converts black-hole radiation conditions into a controlled **non-Hermitian spectral problem**.
- Schwarzschild and RN results show that one spectrum organizes
isolated QNM poles + **rotated continuum (next talk)**.

► Alternative approach to black-hole ringdown as a non-Hermitian eigenproblem.

- The complex-scaled spectral representation

$$R_{\theta}(z) = \sum_{n \in \text{res}} \frac{|\psi_{n,\theta}^R\rangle \langle \psi_{n,\theta}^L|}{z - E_n} + \int_{L_{\theta}} d\lambda \frac{|\psi_{\lambda,\theta}^R\rangle \langle \psi_{\lambda,\theta}^L|}{z - \lambda}.$$

keeps the continuum as part of the response, not as numerical clutter **[Hirose]**.